MAYUREE BINJOLKAR

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Research scientist and ML research lead specializing in reinforcement learning, LLM and VLM post-training, and long-horizon agentic systems, owning research from problem formulation to production deployment. Ph.D. research at the University of Washington developed offline and inverse RL for driver-behavior prediction and adaptive decisions, using VR simulations and human-subject experiments to study driver–automation interactions under distraction and neurocognitive variability.

SELECTED WORK AND RESEARCH EXPERIENCE

Founding Research Scientist, Trapezoid AI

July 2025 – Present

- Led ML R&D for an agentic scene-understanding and identity-resolution pipeline, transforming a SQL workflow into a production AWS Lambda–based system deployed for the first paying customer and expanded to five US locations.
- Built an agent harness coordinating OWLv2, OCR, and selective VLM checks, using intermediate results to choose follow-up checks and retries and validate carrier, truck, and trailer identities.
- Built replay-based evaluation harnesses against labeled ground truth to catch pre-deployment failures, compare model and spatial-feature changes, and guide improvements in identifier accuracy across cameras.
- Post-trained Qwen-family VLMs using LoRA, QLoRA, and DoRA for selective identifier verification, reducing manual review by approximately 10%; quantized Qwen2.5 32B to lower inference cost by approximately 20% per camera event.
- Developed and deployed a reinforcement learning controller that adapts perception and inference configurations across production cameras using detection-performance feedback.

ML Lead & Research Scientist, Meili Technologies

June 2023 – December 2024

- Led ML R&D at Meili, a Cornell Tech spinout, setting technical direction and owning model development, training, and evaluation for multimodal driver monitoring and, following the transition to Artemis, warehouse safety.
- Trained YOLOv8 object detectors and integrated MediaPipe hand tracking into warehouse-safety prototypes to investigate forklift hazards and detect hands outside designated safe zones.
- Developed multimodal camera and electrocardiogram (ECG) models to investigate early prediction of driver medical events; evaluated using in-vehicle data and a driving simulator adapted for use from a hospital bed.
- Applied generative adversarial networks (GANs) and maximum mean discrepancy (MMD) for domain adaptation, evaluating generalization from controlled simulator environments to actual vehicles.
- Built PointNet-based head-pose models from 3D point clouds and trained diffusion models for synthetic-data augmentation to address limited driver-monitoring data.

Researcher, DENSO Seattle Innovation Lab Collaboration

September 2020 – June 2023

University of Washington

- Built CNN–LSTM temporal models for real-time driver-distraction detection across 64 conditions; adapted learned intent models for ADAS with industry engineers and optimized inference with ONNX/TensorRT.
- Collaborated with DENSO researchers to distill ResNet-50 into MobileNetV3 for driver head-pose estimation, using soft-label supervision to enable real-time in-vehicle inference with minimal accuracy loss.

Researcher, Toyota CSRC Collaboration

June 2019 – June 2021

University of Washington

- Developed inverse reinforcement learning frameworks to infer latent decision policies from demonstrations and predict road-user behavior from intersection trajectories (JSM 2020).
- Built an action-conditioned world model over a discrete driving state space, learning next-state dynamics from agent experience; studied policy adaptation across intersection types and under driver distraction.
- Designed VR simulations generating naturalistic behavioral trajectories for RL-based prediction and planning models.
- Collaborated with Toyota CSRC on human-subject experiments examining driver behavior at crosswalks.

Researcher, National Institutes of Health–Funded Study

June 2020 – June 2023

University of Washington

- Applied statistical modeling, unsupervised clustering, and ML feature extraction to VR kinematics and physiological signals to characterize cognitive–motor behavior; findings published in Journal of Neurotrauma, Traffic Injury Prevention, and Transportation Research Part F.

Researcher, Human Factors and Statistical Modeling Lab

June 2019 – June 2023

- Evaluated Batch-Constrained Q-learning (BCQ) and Conservative Q-learning (CQL) for sequential driver decision-making under varying distraction and traffic criticality.
- Studied reward-function changes and state augmentation to capture temporal dependencies; compared policy adaptability and variability across driving conditions.

SELECTED AI RESEARCH

Decision Value of Human Models Under Certified Policy Families and Independent Refitting *2026*

Submitted: ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2027.

- Developed a simulated vehicle-crossing evaluation combining exact reachability certificates, matched controls and independent controller refitting to test the decision value of human-driver information.
- Information improved decisions across all eight refits, but the complete probing strategy underperformed no probing; analyzed 496 state–controller comparisons to identify improvements that persisted or reversed after retraining.

Evaluating Physical Probes for Task Decisions under Hidden Dynamics *2026*

Accepted: PhysWorldAI Workshop, Conference on Neural Information Processing Systems (NeurIPS), 2026.

- Built a MuJoCo/MJX and JAX evaluation protocol separating sensing, state-estimation and planning failures under randomized physical dynamics.
- Across independently seeded 2,048-world studies and 35 probe waveforms, located the main failure before planning: approximately 30% task completion with estimated state versus 100% with true state.

Learning Longitudinal Control Decisions for Autonomous Driving Under Conflicting Collision Risks *2026*

Accepted: UrbanState Workshop, Winter Conference on Applications of Computer Vision (WACV), 2027.

- Learned visual action values from camera and ego-motion histories in CARLA, comparing direct learning with survival-coupled objectives across 5,880 simulation branches and 242 training fits.
- Reduced contact from 78.75% with constant-speed control to 33.50% and increased average progress from 4.95 m to 14.83 m, scoring contact runs as zero; controlled comparisons showed better survival prediction did not improve driving outcomes over direct learning.

Same Threshold, Different Guard: Temporal Force Measurement in Simulated Manipulation *2026*

Submitted: IEEE International Conference on Robotics and Automation (ICRA), 2027.

- Built controlled manipulation experiments in Bullet, MuJoCo and PhysX to isolate how force-measurement windows affect execution, holding dynamics, ordinary feedback and guard response fixed.
- A prespecified 10 ms guard average increased retrieval from 0/128 to 99/128 in fresh Bullet worlds and from 77/128 to 121/128 in a matched PhysX panel; evaluated timestep, force-threshold, grip-strength and timing sensitivity.

Answer-Controlled Probing of Spatial Information in Latent Visual Reasoning *2026*

Accepted: Latent Visual Reasoning Workshop, Winter Conference on Applications of Computer Vision (WACV), 2027.

- Probed spatial information in frozen LatenRn-3B-RL representations across 200 scene pairs using answer-preserving edits, pre-answer representations and coordinate probes.
- Internal spatial-relation predictions reached 94.24% accuracy versus 60.40% for the strongest answer/confidence control; baseline comparisons limited evidence that internal reasoning caused the advantage.

Auditing the Monitorability of Reasoning Models Under Quantization and Thinking Budgets *2026*

Efficient Reasoning Workshop, Conference on Language Modeling (COLM), 2026.

- Audited reasoning-model faithfulness, safety and monitorability across weight precisions and reasoning-token budgets, covering BF16, INT8, INT4-AWQ and 2-bit GGUF.
- Built a vLLM/llama.cpp evaluation harness over 9,840 generations with controlled token limits, two LLM judges, blind human adjudication and bootstrap/regression analysis.

TECHNICAL SKILLS

- **RL for Reasoning and Agents:** RLHF, RLVR, PPO, GRPO, DAPO, GSPO, reward modeling, process/outcome rewards, long-horizon credit assignment and exploration.
- **RL for Control and Adaptation:** Offline and inverse RL, SAC, TD3, CQL, IQL, BCQ, model-based planning, meta-RL and policy adaptation.
- **Agent Evaluation and Harness Design:** Evaluation harnesses, trajectory/outcome assessment, replay-based regression testing, LLM-as-judge, ablations, representation probing and statistical inference.
- **Model Adaptation and Optimization:** SFT, LoRA, QLoRA, DoRA, preference optimization, distillation, domain adaptation, quantization, GANs and diffusion-based augmentation.
- **Multimodal and Physical AI:** VLMs, sensor fusion, spatial reasoning, temporal modeling, action-conditioned world models, learned dynamics and state estimation.
- **Frameworks, Simulation and Deployment:** PyTorch, JAX/MJX, TensorFlow, Transformers, MuJoCo, PyBullet, PhysX, vLLM, llama.cpp, ONNX/TensorRT; Python, C++, SQL, R, Docker, AWS and GCP.

EDUCATION

- **Ph.D., Data Science**, University of Washington — *Offline RL, Inverse RL, and sequential decision-making* *2024*
- **M.S., Computer Science and Systems Engineering**, University of Washington
- **M.S., Transportation Engineering**, University of Washington
- **B.S., Engineering**, Technion – Israel Institute of Technology

SELECTED PUBLICATIONS & PRESENTATIONS

- **Binjolkar, M.** Learning Longitudinal Control Decisions for Autonomous Driving Under Conflicting Collision Risks. UrbanState Workshop, WACV, 2027 (accepted).
- **Binjolkar, M.** Answer-Controlled Probing of Spatial Information in Latent Visual Reasoning. Latent Visual Reasoning Workshop, WACV, 2027 (accepted).
- **Binjolkar, M.** Evaluating Physical Probes for Task Decisions under Hidden Dynamics. PhysWorldAI Workshop, NeurIPS, 2026 (accepted).
- Binjolkar, Manjaree, and **Binjolkar, Mayuree**. Where Does Neural Ocean Emulator Error Live? A Depth-Resolved Decomposition. FMTS Workshop, NeurIPS, 2026 (accepted).
- **Binjolkar, M.** Auditing the Monitorability of Reasoning Models Under Quantization and Thinking Budgets. Efficient Reasoning Workshop, COLM, 2026.
- **Binjolkar, M.**, Tu, Y. Adaptive Decision-Making in Human-AI Teaming for Dynamic Driving Environments. INFORMS Annual Meeting, 2024.
- Saraiya, V., **Binjolkar, M.** Optimizing Energy Consumption in AI Training: A Reinforcement Learning Approach. INFORMS Annual Meeting, 2024.
- Tu, Y., **Binjolkar, M.** Enhancing electric vehicle charging demand prediction with graph neural networks. INFORMS Annual Meeting, 2024.
- **Binjolkar, M.**, et al. The functional capabilities of young drivers and implications for fitness-to-drive. Transportation Research Part F, 2024.
- Patrick, K. E., Kroshus, E., Boyle, L. N., Wang, J., **Binjolkar, M.**, et al. Driving characteristics of young adults prior to and following concussion. Traffic Injury Prevention, 2024.
- **Binjolkar, M.**, Park, S., Kim, S. Explainability as a Path to Foster Trust in Autonomous Driving Decision-Making. Inference and Decision Making for Autonomous Vehicles Workshop, Robotics: Science and Systems, 2023.
- Rivara, F. P., Ebel, B. E., **Binjolkar, M.**, et al. Cognitive Impairment and Driving Skills in Youth After Concussion. Journal of Neurotrauma, 2023.
- **Binjolkar, M.**, Sosnovskaya, Y. Augmenting Driver Decision-Making Using Meta-Inverse Reinforcement Learning. Poster, WiML, NeurIPS, 2022.
- **Binjolkar, M.**, Boyle, L. N. Capturing Driver Status Using Naturalistic Driving Study Data: Implications for Assessing Health and Safety. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2022.
- **Binjolkar, M.**, Ng Boyle, L. Using Inverse Reinforcement Learning to Predict the Impact of Distraction on Future Driver Performance. Joint Statistical Meeting, 2020.
- Mittal, V., Garg, N., Wagenmans, Y., **Binjolkar, M.**, et al. Viruses, Visualization, and Validation: Interactive Mining of COVID-19 Literature. ISMB, 2020 (abstract accepted for oral presentation).